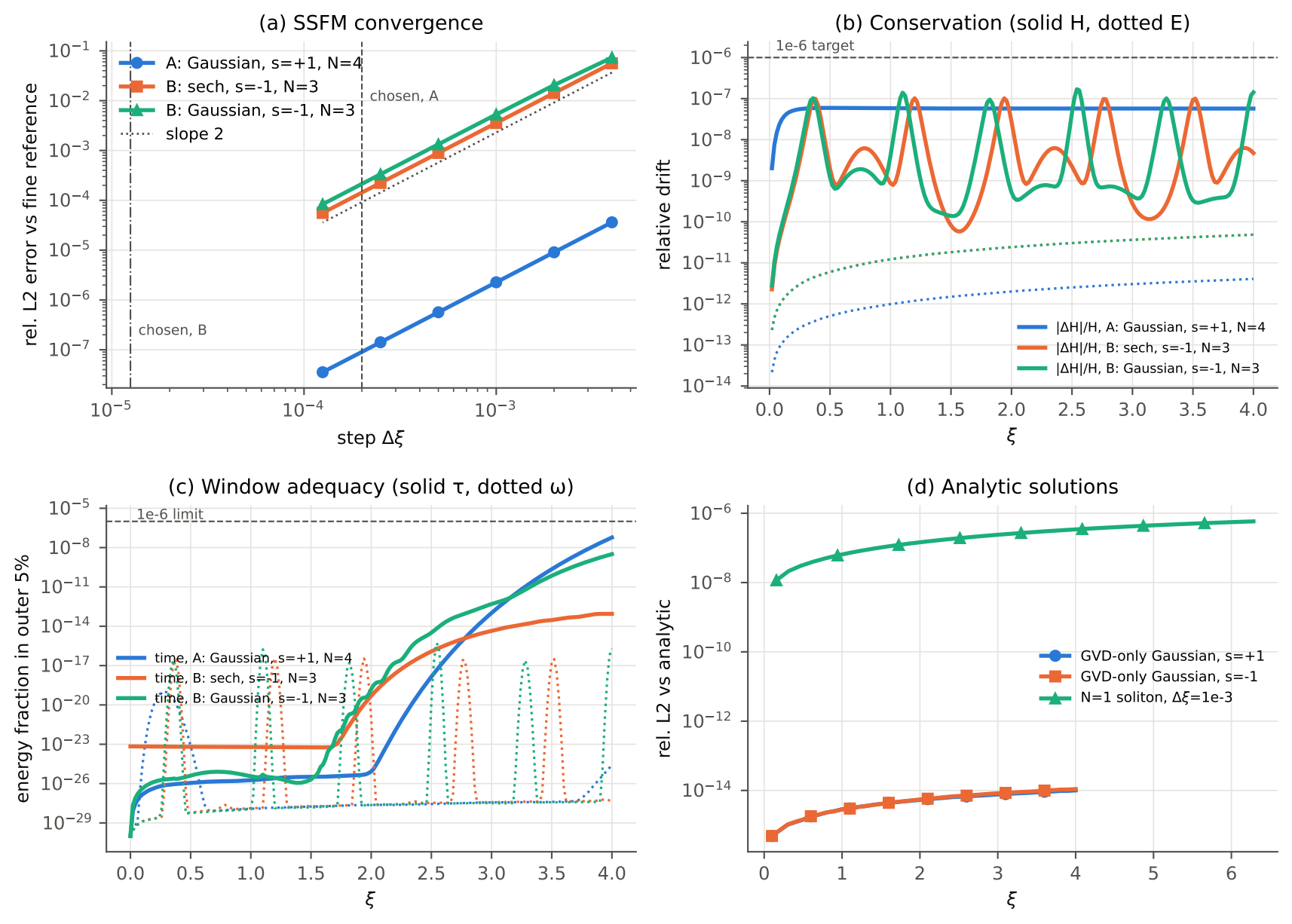
(a) SSFM convergence
A: Gaussian, s=+1, N=4
B: sech, s=-1, N=3
B: Gaussian, s=-1, N=3
slope 2
chosen, A
chosen, B
rel. L2 error vs fine reference
step Δξ
(b) Conservation (solid H, dotted E)
1e-6 target
relative drift
|ΔH|/H, A: Gaussian, s=+1, N=4
|ΔH|/H, B: sech, s=-1, N=3
|ΔH|/H, B: Gaussian, s=-1, N=3
ξ
(c) Window adequacy (solid τ, dotted ω)
1e-6 limit
energy fraction in outer 5%
time, A: Gaussian, s=+1, N=4
time, B: sech, s=-1, N=3
time, B: Gaussian, s=-1, N=3
ξ
(d) Analytic solutions
rel. L2 vs analytic
GVD-only Gaussian, s=+1
GVD-only Gaussian, s=-1
N=1 soliton, Δξ=1e-3
ξ